

Nanomand

Nanomand Team - BA468

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Nanomand's Context

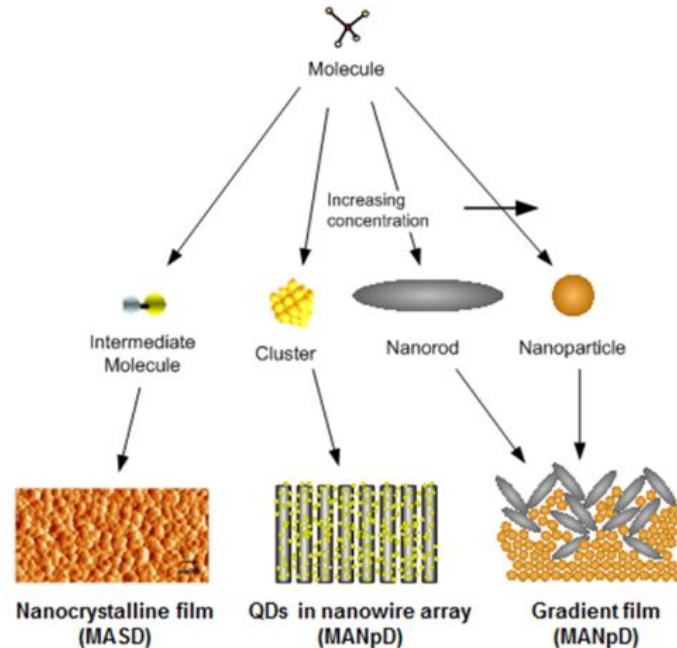
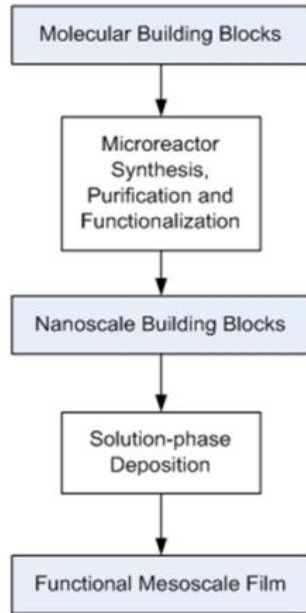
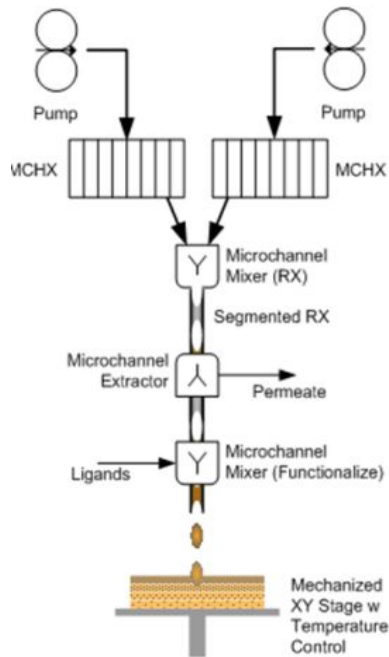




Nanomand

- **Company:** Nanomand
- **Location:** Corvallis, Oregon
- **Type:** Materials Technology Startup (OSU Advantage Accelerator)
- **Company Stage:**
 - Early-stage research startup
- **Key Advantage**
 - Patented material integration process
 - Experienced University Research Team
- **Core Technology**
 - Proprietary nanoparticle reactor
 - Embeds nanoparticles into metals and other materials
- **What This Enables**
 - Higher hardness and thermal conductivity
 - Materials beyond limits of traditional alloys

Refresher - Functional Nanomaterials on Demand





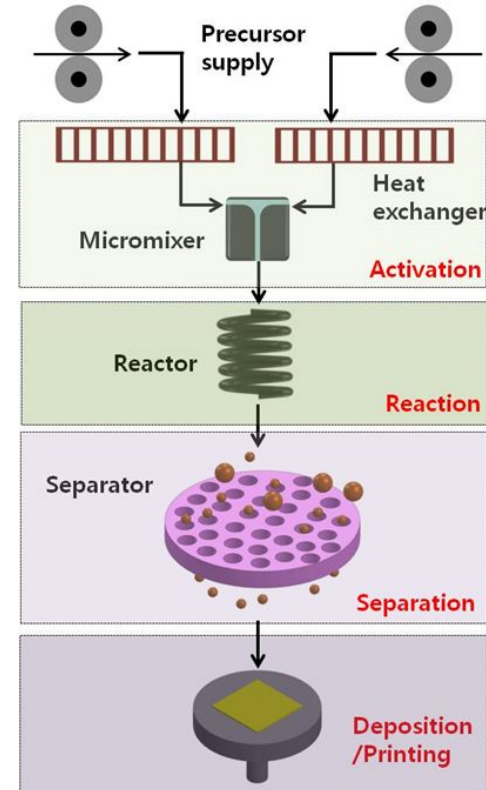
Technology

Microreactor -> Material Synthesis

Defined Microstructures

Nanoparticle Application at Points of Interest

Just-In-Time Manufacturing



Customer Discovery





AddiTec

Application of Nanomand

Integration of reactors into WAAM printheads

Value provided by Nanomand

Real-Time Material Property Tuning

Key Takeaway

Nanomand can extend AM beyond conventional limits.



Industries of Interest

Semiconductor

- Heat Management
- Material Precision

Tool Coating

- Durability
- Hardness
- Hot Hardness



Experts

- **Amy M. Elliott** at Oak Ridge National Laboratory
- **Chinedum Okwudire** at Ulendo, a 3D printing company.
- **Travis Mayberry** at Divergent
- **Richard Wirz** at Boeing & Oregon State University
- **Bahman Abbasi** at Oregon State University
- **Benjamin Adam** at Oregon State University
- **Bob Kaplan**, Oregon State University Alumnus
- **Scott Campbell**, Oregon State University



Scott Campbell Interview Results

- Positively received the idea of Nanomand
- Most valuable traits were:
 - 1) Durability (Staying hard for longer)
 - 2) Physical Hardness
 - 3) Hot Hardness (Softer as they get Hotter)
 - 4) Thermal Conductivity (Injection Molding only)
- Titanium was currently the best coating material for machined tools
- Beryllium Copper is currently is the best material for injection molds
- Accessible Lab Testing at OSU



Target Businesses & Contacts

- **Intel:** Marcos Hernandez (Module Engineer)
 - Semiconductor Industry
- **Broadcom:** Derui Kong (Principle Design Engineer)
 - Semiconductor Industry
- **Lockheed Martin:** Nathan Richardson (Associate Fellow)
 - Tool Coating (Aerospace) Industry
- **Boeing:** Will Newberry (Process Engineer)
 - Tool Coating (Aerospace) Industry



Interview Structure

Main Topics:

- Value Proposition
 - Do they find value in the concept of Nanomand?
- Customer Problem
 - What specific problem does this material solve that costs customers money today?
- Performance Impact
 - What measurable performance improvement do we deliver, and how large is it?
- Buyer & Decision Path
 - Who would be willing and able to adopt this?
- Pilot & Testing Path
 - What is the fastest way a customer could test or pilot this in their existing process?
- Early-Adopter Industries
 - What industry can adopt this without long certification or regulatory delays?

Market Research





Research Efforts

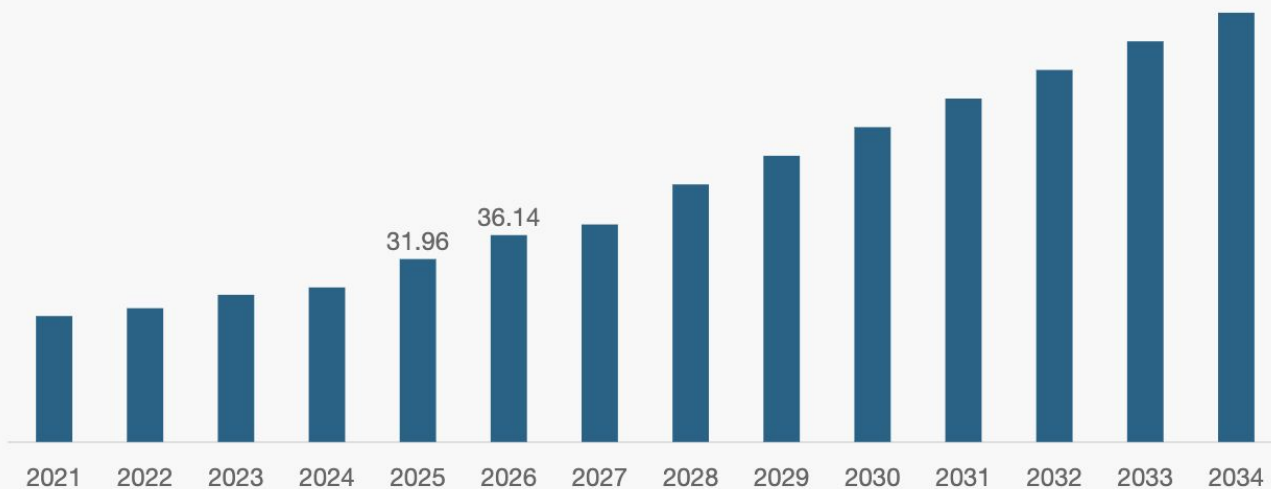
Researched Nanoparticle Industry Information:

1. Looked for general global market size and growth current status / projections
 - a. Attempted to narrow results to North American nanoparticle industry
2. Looked for the industry breakdown for where Nanomand can be applied
 - a. Ignored things such as health care, or the semiconductor silicon manufacturing process
3. Examined market trends that affect the North American nanoparticle industry
 - a. Nanomand being a US based company will have a easier time selling solutions to the North American Market initially



Nanotechnology Market

North America Nanotechnology Market Size, 2021-2034 (USD Billion)

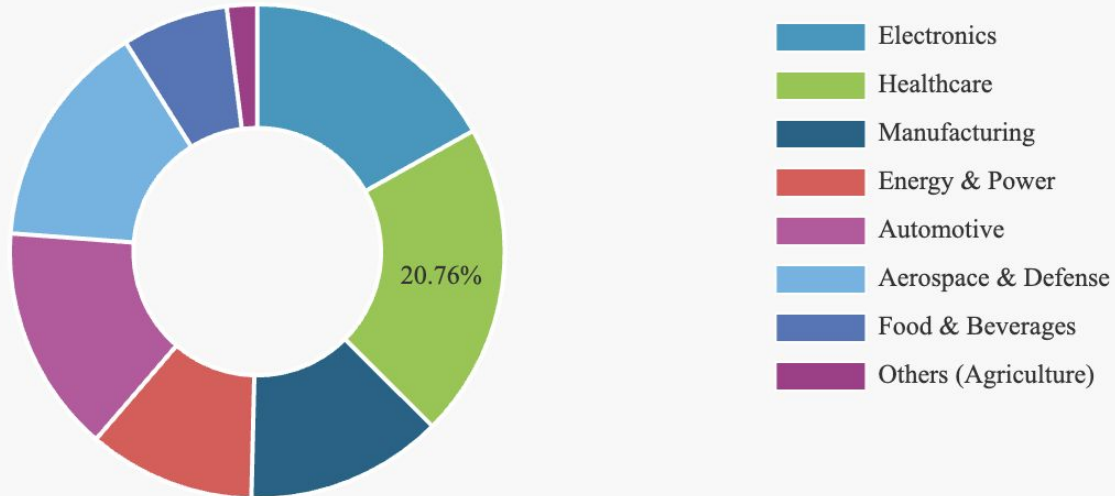


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Nanotechnology Market

Global Nanotechnology Market Share, By Industry, 2026



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Trends to Opportunities vs. Demand

Trends to Opportunities			Trends to Demand			
Description			"TAM"	"SAM"	"SOM"	
Mega Trend	Effects	Opportunities	Total Addressable Market	Serviceable Addressable Market	Serviceable Obtainable Market	Percent Annual Growth Rate
General AI Boom	There are many AI Data Centers being built due to companies competing for Compute power	Nanomand can help develop high powered cooling solutions	\$ 12,000,000,000	44%	8%	
				\$ 5,220,000,000	\$ 417,600,000	21.2%
High Efficiency Tools for cutting, drilling and milling	Increased demand for tools that can support efficiency/longevity of cutting, drilling, and mining efforts.	Nanomand can create high quality tool coatings that provide increased Life Span or Performance	\$ 5,150,000,000	65%	12%	
				\$ 3,347,500,000	\$ 401,700,000	5.2%

Trends to Opportunities Cont.

Trends to Opportunities		
Mega Trend	Effects	Opportunities
Battery anode breakthroughs using silicon nanowires for EV range	This can accelerate the shift towards more efficient EVs with improved battery performance, reduction in charging times, and increased longevity.	Nanomand can develop scalable materials that could boost EV range, charging performance, and overall longevity.
United States CHIPS Act support domestic nanosensor fabs	Semiconductor manufacturing will see increased demand as nanosensor Fabs (fabrication facilities) receive more government support, and thus additional funding.	Nanomand can leverage the increased government funding for nanosensor fabs.
Rapid adoption of quantum-dot displays by Asian TV OEMs	High adoption of nanometer-scale, semiconductor crystals will result in increased demand for mineral based nanoparticle application.	Nanomand could leverage the increased revenue and thus R&D expenditure from Asian TV OEM companies to insert themselves into their production line.



Total Market Trends

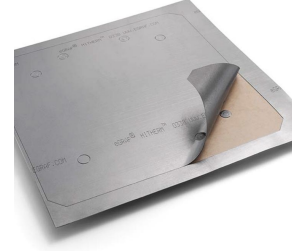
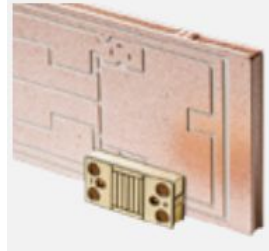
<i>"TAM"</i>	<i>"SAM"</i>	<i>"SOM"</i>	Annual % Growth
\$ 105 B	\$ 32 B	\$ 3.5 B	15.7%

Competitor Research



Competitors - Cooling Materials

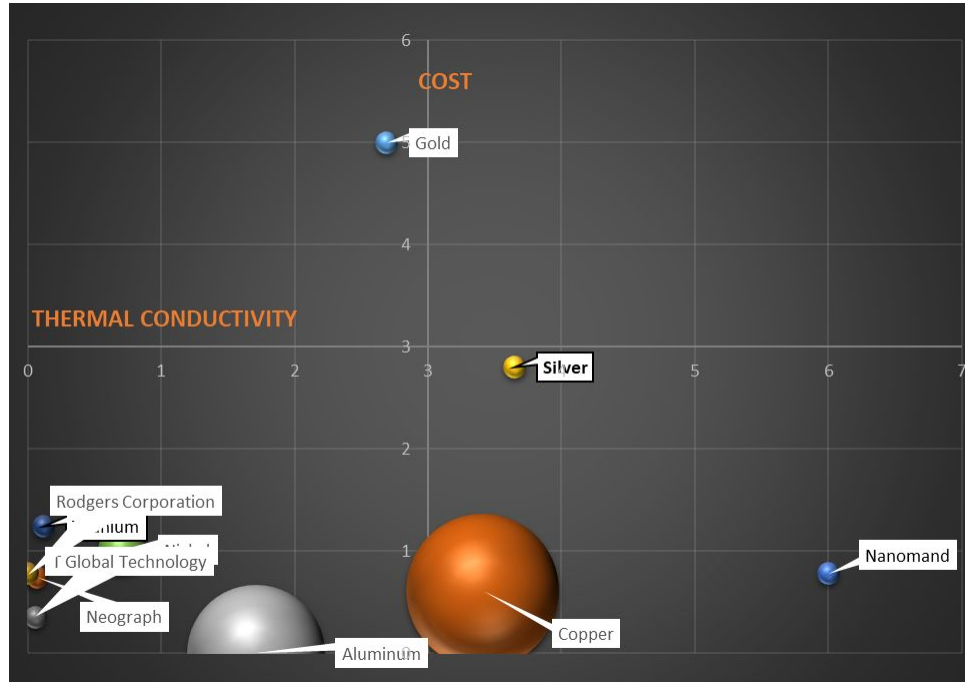
Company Name	Your Client's Company Name	Primary Competitor's Name	Competitor's Name	Competitor's Name
Product Name	Nanomand	Rogers Corporation's Curamik cooling solutions	Neograf's eGraf SpreaderShield Heat Spreaders	T Global Technologies' T68 Synthetic Graphite Sheet
Unit Price (per kg)	\$14 (copper based)	\$14 (copper based)	\$13	\$6
Thermal Conductivity (W/mK)	700 (through-plane)	400 (through-plane)	350-600 (in-plane) 3 (through-plane)	1500 (in-plane) 5 (through-plane)



Competitors - Cross Plane



Competitors - Throughplane



Commercialization





Industries to Target:

Semiconductors		Tool Coatings	
Pro:	Con:	Pro:	Con:
Very large and fast-growing market	Extremely difficult market entry	Small parts make prototyping easier	Highly competitive industry
Clear performance advantage opportunity	Very large order Volumes	Faster adoption cycles	Lower economic upside per product
High value per improvement	High technical and infrastructure barriers	Accessible testing infrastructure	Performance improvement must be significant



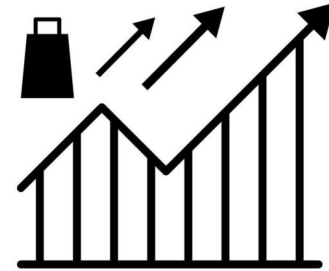
Business Model Pathways

- **Custom Material R&D**
 - Work with companies
- **Advanced Coating Supplier**
 - Provide to manufacturers
- **Licensing**
 - Companies integrate with their own products
- **Productizing**
 - Develop productized solutions for industries



Promotional Activities

- Industry Trade Shows
- Engineering Newsletters
- Technical Articles
- Conferences



Marketing Approaches



Channels

Selling Locations

- University Labs / Partnerships
- Trade Shows
 - SEMICON West
 - Advanced Materials Show
- Large Scale Material Supply
- Online - Once productized

Distribution Channels

- AddiTec
- Digital Storefronts / Marketplaces
- Directly B2B
- Licencing



Major Accessories

- Raw Input Materials
- AddiTec 3D Printing Technology
- Engineering Labs at OSU



Summary

Customer Discovery

- Intel / Broadcom (Semi)
- Lockheed / Boeing (Tool)

Market Research

- Semiconductor
- Tool Coating

Competitor Research

- Higher Relative Cost
- More Thermal Conductivity

Commercialization

- Coating Supplier (+ R&D)
- Personal Selling